

Brian Lee

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US Citizen – eligible for security clearance

PROFESSIONAL SUMMARY

Software engineer with 2+ years building AI-enabled scientific computing and simulation systems for national security applications, including AI integration into simulation frameworks, ML evaluation tooling, and GPU-accelerated training pipelines. Experienced developing modular OOP systems in Python, deploying on cluster compute (SLURM/CUDA/MPI), and collaborating across multidisciplinary research teams. Multiple peer-reviewed IEEE publications.

EXPERIENCE

Virginia Tech National Security Institute Research Software Engineer

Arlington, VA

Aug. 2023 – Aug. 2025

- **AI/simulation integration:** Implemented and integrated computer vision pipelines into MIMIK, an open-source mission engineering tool for kill web simulation, to emulate AI agent perception in simulating decision modeling.
- **ML testing software:** Led development of CODEX, an open-source Python tool implementing combinatorial coverage for ML data analysis. Refactored ~85% of the previous codebase from C into a modular OOP design in Python, built data analytics/experimentation modules, and applied the tool to cross-domain experiments.
- **Deep learning experimentation:** Trained, fine-tuned, and debugged deep-learning training pipelines that enabled experiments on dataset bias on SLURM-managed HPC cluster (Tinkercliffs) using CUDA-distributed compute.

Virginia Tech National Security Institute Research Intern

Arlington, VA

May 2022 – Aug. 2022

- **Data pipeline engineering:** Built an image pipeline in Python that scraped and preprocessed **150k+** images from Instagram at ~120 images/min.
- **Generative model experiments:** Designed controlled multi-GPU StyleGAN3 training experiments comparing transfer learning to from-scratch baselines on low-resource datasets, achieving **1.5×** faster FID convergence.

SKILLS

Languages: Python; C/C++; Java; MATLAB/Simulink

ML/DL: PyTorch; TensorFlow; NumPy; Pandas; scikit-learn; StyleGAN3; YOLO; CUDA

HPC / Parallel: Cluster computing (VTARC's Tinkercliffs); Linux/UNIX; Multi-GPU; Batch scripting; MPI; CMake

Software engineering practices: Git; Confluence/Jira; CI/CD pipelining; agile development; unit testing

PROJECTS

MIMIK (Mission Engineering Tool) | Open-source · github.com/vtnsi/mimik

Aug. 2024 - Jun. 2025

- Developed ML software components for a mission engineering tool that uses graph-based modeling to simulate decision-making paradigms for military kill webs.
- Designed and integrated end-to-end computer vision pipelines in Python (YOLOv8) covering image retrieval, preprocessing, dataset reformatting, and model training, into the simulation framework.
- Integrated object detection model evaluation outputs into kill web decision-making to compare kill web success between algorithmic and AI-enabled decision-making approaches.

CODEX (Coverage of Data Explorer) | Open-source · github.com/vtnsi/codex

Aug. 2023 – Jun. 2025

- Built the analytics/visualization module that uses Matplotlib and Seaborn to produce metrics reports, coverage and skew plots, and coverage-performance figures used by analysts to study ML datasets across disciplines.
- Integrated training pipelines for select ML models (incl. YOLOv8), enabling data-to-performance testing capabilities.

PUBLICATIONS

- Mhatre, S., Lee, B., et al. "Coverage Metrics for Detecting Bias in Facial Recognition Datasets," in *2025 IEEE Conf. on AI*, 2025. Presented May 2025.
- Lanus, E., Lee, B., et al. "Coverage for Identifying Critical Metadata in Machine Learning Operating Envelopes." *2024 IEEE International Conference on Software Testing, V&V Workshops (ICSTW)*, Toronto, ON, CA, 2024.

EDUCATION

Virginia Tech
B.S. in Computational Modeling and Data Analytics
Minors in Computer Science and Mathematics

Blacksburg, Virginia
Aug. 2019 - May 2023